The Shackleton rover: An Expedition to the Lunar South Pole

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Investment Plan

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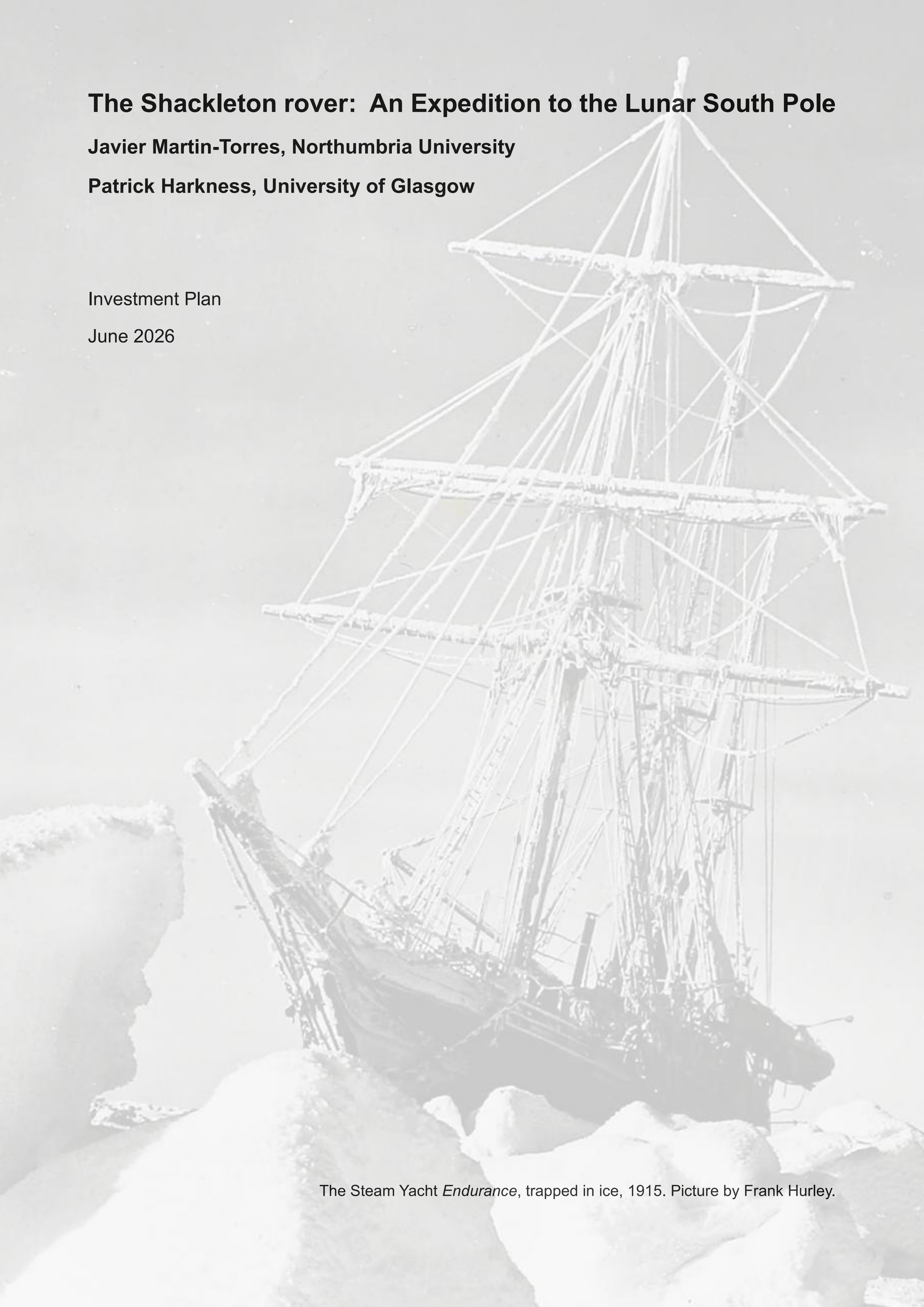

The Steam Yacht *Endurance*, trapped in ice, 1915. Picture by Frank Hurley.

1. Foreword by our concept sponsor, The Hon. Alexandra Shackleton:
"My grandfather, Sir Ernest Shackleton, listed the qualities required for a polar explorer thus: optimism, patience, imagination, and courage. I believe that he would recognise these qualities as motivating this proposal, and I am therefore delighted to commend it to you."

2. Outline: Sir Ernest Shackleton's famous *Endurance* expedition has inspired ambition, leadership, and moral optimism for more than a century.

Today, a similar opportunity exists at the Moon's south pole. We have the scientific expertise, engineering capability and operational experience required achieve an important milestone in planetary science: first contact with water-ice on the surface of the moon.

The *Shackleton* rover, described at shackletonrover.co.uk, is a proposal for a British lunar rover to accomplish this task. This mission will draw on our heritage of exploration, as our name makes clear. We will also carry a fragment of the *RRS Discovery* to the moon, just as NASA's *Perseverance* rover carried a piece of the *Wright Flyer* to Mars.

We must move quickly, because we have a short window of opportunity to achieve this globally-significant first. Many nations are building lunar rovers, targeting flight in the next few years. The search for ice is ubiquitous, but the ice itself is likely underground. Until American astronauts arrive, the exploration of the subsurface will always be difficult because drilling holes in space is extraordinarily difficult: in low gravity, there is low weight, and without weight being applied no drillbit can ever make any progress.

However, the UK has unique technologies which can fluidise soil and allow drilling at much lower weights than would otherwise be possible. We also have a ground station at Goonhilly which is able to communicate with assets on the Moon. There are few nations with a better chance of securing this achievement at a reasonable price. We have a hugely capable team, with wide experience across a range of instruments that have already flown to both Moon and Mars. We are ready to fly our own lander, and operate our own mission.

We believe that this mission has the potential to captivate the nation, building on Shackleton's spirit of optimism, competence, and perseverance. The UK needs to recover its national morale, and we will create an inspirational showcase that will attract more young people to STEM. We will build world leaders in space exploration, just in time to contribute to our journey to the Moon, Mars, and beyond.

Finally, we will carry out this mission at the centre of an international team. To ensure 24-hour operations on the moon, we will have driver teams in Canada and Australia, covering each others' unsociable hours. This international element is deliberate, because we want our mission to showcase international partnership as a mechanism for lunar exploration. First contact with space resources will be a convincing moral statement of intent, but we will use this to underline our commitment to the idea of a new global commons on the Moon, as is protected by the Antarctic Treaty System on Earth. Helping to create a new Antarctica, in space, with all the protections and achievements of the old one, will be a very worthwhile legacy of our endeavour.

3. Costs: Shackleton rover could be built and operated for two weeks for a total investment of around £22m. A CLPS delivery to the Moon may be on the order of £3m.

COST SUMMARY BY CATEGORY			
Category	**Cost (£)**	**% of Total**	**Notes / Heritage**
Rover Platform (Structure, Mobility, Power, Thermal, Comms, Avionics)	£6,500,000	29.5%	*Northumbria, Leicester, Open Univ?*
Scientific Payloads (Mass Spectrometer Suite + Sonicated Penetrator)	£7,000,000	31.8%	*Open Univ (MS heritage from Peregrine); Glasgow/Leeds/BAS (penetrator)*
Assembly, Integration, Test & Verification (AITV)	£2,200,000	10.0%	*Glasgow dirty-vacuum, Leeds simulant tests, full environmental qual*
Project Management, Systems Engineering & Assurance	£1,800,000	8.2%	*Lead: University of Glasgow / consortium coordination*
Ground Segment, Software, Launch Integration & Ops Preparation	£2,700,000	12.3%	*Goonhilly, international partner ops ???? (Canada/Japan/Australia)*
Mission Operations, Data Analysis & Science Support	£1,000,000	4.5%	*14-day 24/7 ops with partner rotation; quick-look science*
Education, Training, Public Engagement & Outreach	£500,000	2.3%	*PhD cohort aligned to 4-year programme; STEM inspiration*
Contingency & Risk Reserve	£300,000	1.4%	*~1.4% reserve; higher TRL risk on novel penetrator mitigated by testing*
TOTAL ESTIMATED MISSION COST	**£22,000,000**	**100.0%**	**Benchmark: Australian Roo-ver A$42M (~£22M) for equivalent scope**

WBS	Category / Element	Cost (£)	%	Primary UK Lead / Heritage & Notes
1.0	PROJECT MANAGEMENT, SYSTEMS ENGINEERING & ASSURANCE	£1,800,000		University of Glasgow (lead); consortium of all signatories
1.1	Project Management & Control	£600,000		Full lifecycle PM, risk registers, schedule/cost control
1.2	Systems Engineering & Interface Control	£700,000		Requirements flow-down, ICDs with CLPS provider, verification planning
1.3	Quality Assurance, Safety & Reviews	£500,000		PDR/CDR/TRR/FRR; planetary protection compliance
2.0	ROVER PLATFORM	£6,500,000		Northumbria, Leicester, Open?
2.1	Structure & Chassis (0.4m x 0.15m sides)	£1,200,000		Lightweight composite/metal; sized for 0.25m² side solar panels
2.2	Mobility System (4 wheels, skid-steer, gearmotors)	£1,800,000		~750g per corner; low-mass wheels for lunar regolith
2.3	Power System (Solar + Battery)	£1,100,000		0.25m² panels (~40W), 50Wh Li-ion ; 3kg/m² & 200Wh/kg assumptions
2.4	Thermal Control (Insulation + PCM + RHU)	£700,000		Phase-change material for cold-soak sorties to shadowed regions
2.5	Communications (S-band, deployable mast, patch antenna)	£900,000		Directional patch on yaw-controlled mast (Astromast/Rolatube heritage)
2.6	Avionics, Onboard Computer & Harness	£800,000		Nominal 2.5% mass allocation; radiation-tolerant COTS where possible
3.0	SCIENTIFIC PAYLOADS	£7,000,000		Core science return
3.1	Mass Spectrometer Suite (TG-EGA)	£4,200,000		Open University ??; adaptation of Exospheric MS / PITMS (Peregrine heritage). 4kg. Stepped heating, ion trap, evolved gas analysis for H_2O discrimination (adsorbed / pore / cement) + co-volatiles
3.2	Sonicated Penetrator System	£2,800,000		Glasgow?? (ultrasonics lead), Leeds?? (granular modelling), BAS ??(icy drilling). 3kg to 20cm depth. Standing-wave fluidisation reduces penetration force by ~10x in granular media. TRL ~4.5; thermal-vac testing critical. Includes heaters, thermistors, ultrasonic transducer, penetration sensors
4.0	ASSEMBLY, INTEGRATION, TEST & VERIFICATION	£2,200,000		UK national facilities
4.1	Integration & Functional Testing	£800,000		Cleanroom assembly, functional checkout, software-in-the-loop
4.2	Environmental Testing (Vibe, TVAC, EMI/EMC)	£700,000		Standard qual for lunar environment
4.3	Regolith & Penetrator Performance Testing	£700,000		Glasgow dirty-vacuum chamber (one of Europe's largest); Leeds granular flow models; icy-regolith analogues; calibration of force reduction & volatile release
5.0	GROUND SEGMENT, SOFTWARE & LAUNCH INTEGRATION	£2,700,000		
5.1	Flight Software & Autonomy	£900,000		Path planning from mast imagery, obstacle avoidance, autonomous sortie execution
5.2	Ground Control Software & UK Ground Station	£700,000		Goonhilly dedicated command capability; full-spectrum ops demo
5.3	CLPS Integration, ICDs, Mission Assurance	£700,000		Payload accommodation, mechanical/electrical interfaces, safety reviews
5.4	International Partner Coordination	£400,000		Ops hand-over with Canadian/Japanese/Australian teams for 24h coverage
6.0	MISSION OPERATIONS & DATA ANALYSIS	£1,000,000		
6.1	Surface Operations (14 days, 24/7)	£600,000		Real-time commanding, health monitoring, sortie planning; partner-supported shifts
6.2	Data Processing, Archiving & Science Quick-Look	£400,000		Volatile profiles, thermal properties, regolith strength; public data release
7.0	EDUCATION, TRAINING & OUTREACH	£500,000		Critical for skills pipeline
7.1	PhD Studentships & Early-Career Training	£350,000		Aligned to 4-year timeline; hands-on hardware + ops experience
7.2	Public Engagement & STEM Inspiration	£150,000		National narrative (Shackleton spirit); school/university outreach; media
8.0	CONTINGENCY & RISK RESERVE	£300,000		~1.4% of total; applied to highest-risk elements (penetrator TRL advancement)
	GRAND TOTAL	**£22,000,000**	**100%**	

The profile follows a typical project curve.

Category	Year 1 (Concept/Feasibility)	Year 2 (Prelim & Detailed Design)	Year 3 (Build, IATV Peak)	Year 4 (Final Test, Launch, Ops)	TOTAL
Project Management, SE & Assurance	£450,000	£600,000	£500,000	£250,000	**£1,800,000**
Rover Platform	£800,000	£2,000,000	£2,800,000	£900,000	**£6,500,000**
Scientific Payloads (MS + Penetrator)	£1,200,000	£2,500,000	£2,500,000	£800,000	**£7,000,000**
Assembly, Integration, Test & Verification	£200,000	£400,000	£1,200,000	£400,000	**£2,200,000**
Ground Segment, Software & Launch Integration	£300,000	£700,000	£900,000	£800,000	**£2,700,000**
Mission Operations & Data Analysis	£0	£0	£200,000	£800,000	**£1,000,000**
Education, Training & Outreach	£150,000	£150,000	£100,000	£100,000	**£500,000**
Contingency & Risk Reserve	£50,000	£100,000	£100,000	£50,000	**£300,000**
ANNUAL TOTAL	**£3,150,000**	**£6,450,000**	**£8,300,000**	**£4,100,000**	**£22,000,000**
% of Total Mission Cost	**14.3%**	**29.3%**	**37.7%**	**18.6%**	**100%**

4. Funding requirement: We believe that we may be able to leverage approximately £2m from the UK Space Agency NSIP programme, and approximately £1m from the International Bilateral Fund. We further believe that the strategic partnerships with Australia and Canada would be within the scope of the Foreign, Commonwealth, and Development Office Tactical Fund, which may provide another £2m.

We are therefore seeking to raise £20m from philanthropic, industry, and charitable sources.

The purpose of this funding will be to support a mission that has been calibrated to be something more than a minimum viable product: we will, with success, be the first to make physical contact with water-ice on the moon. However, this baseline-plus approach makes sense: we have some UK technologies that have been designed specifically for this exploration task. We can aim higher than most without incurring excessive risk.

Therefore, although our first ambition is for inspiration and achievement, our secondary task is technology demonstration. The cis-lunar economy is estimated to be worth $5bn, in Europe alone, over the next decade. We have an opportunity to make the UK a go-to partner in pursuit of the space resources that will underpin this economy.

5. Next steps: We have sought to identify a mission that is scientifically significant, financially realistic, and which has the right combination of historical resonance and technical ambition to inspire a generation of young people.

We have assembled an engineering team which is well-placed to deliver this ambition, but we do not yet have the complete coalition required to deliver it. We are therefore looking for partners, supporters, critics, advisors, and investors who are willing to engage with the idea and help us determine the best route forward. You may have expertise in finance, industry, philanthropy, government, education, spaceflight, or simply experience in bringing ambitious projects to life. All of these perspectives would be valuable.

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